BISTRO-XAI

Attention Analysis & Feature Selection Report

Korean CPI (YoY) Forecasting with BISTRO Foundation Model

Model: BISTRO (91M params, 12 layers, 12 heads, fine-tuned on BIS 4,925 series)

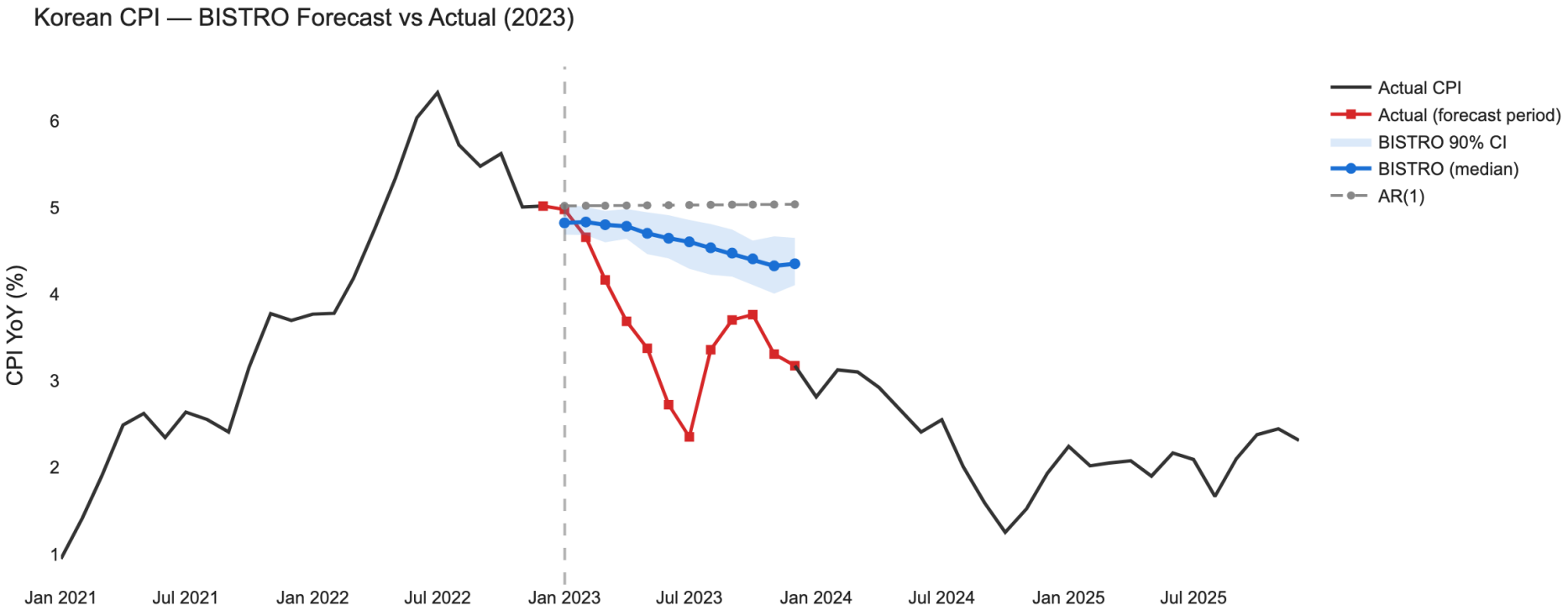
Target: CPI_KR_YoY | Covariates: 10 selected from 28 candidates

Context: ~120 patches (~10 years) | Forecast: 2023-01 ~ 2023-12

Data mode: DAILY

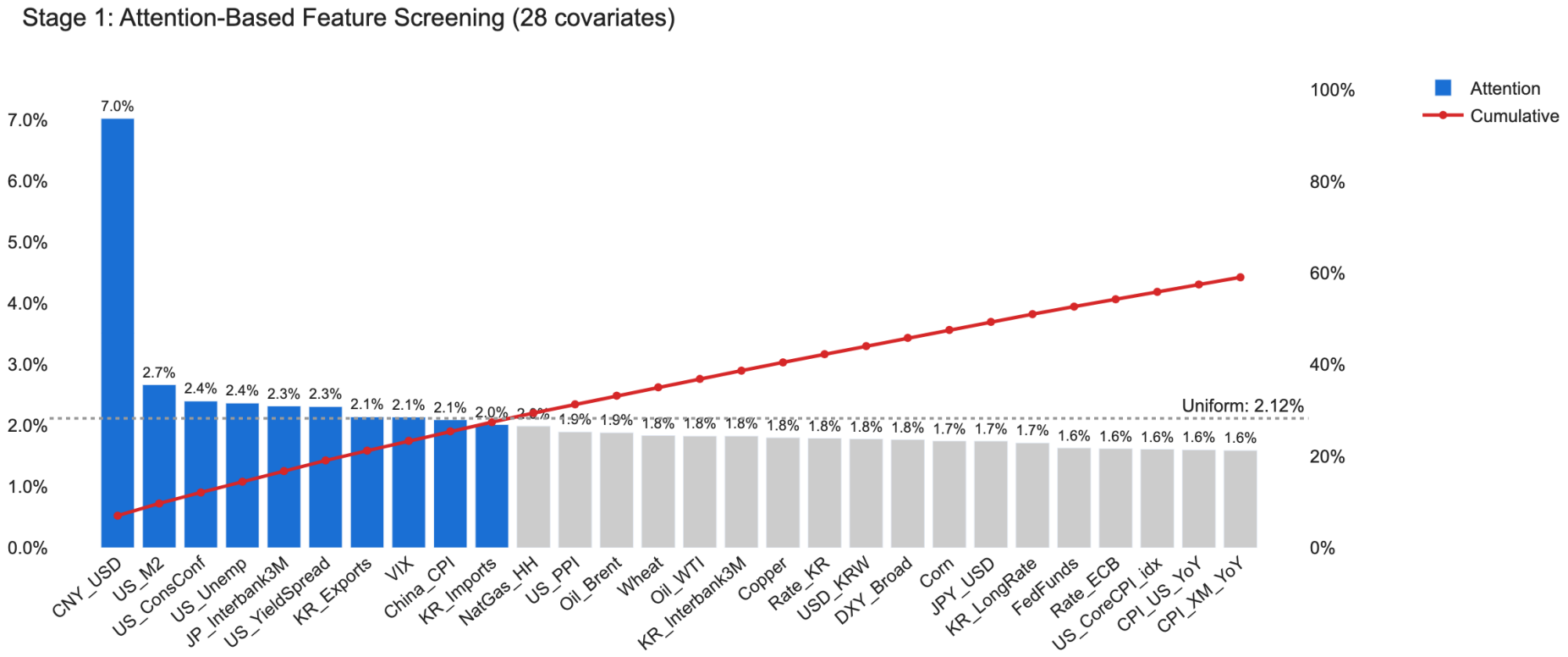
1. Forecast Results

BISTRO median forecast vs Actual CPI (2021-2025). Red squares = actual CPI during forecast period (2023). The 2023 disinflation trend was not captured by the model -- BISTRO predicted persistent inflation while actual CPI declined steadily.



2. Feature Selection (Stage 1)

Attention-based screening: 10 variables selected from 28 covariates. Uniform share threshold: 2.12%. Blue bars = selected for Stage 2, gray = filtered out. Variables above the uniform line receive above-average attention from CPI.



3. Cross-Variate Attention Matrix

Stage 2 (Layer 12, last layer). Self-attention (CPI->CPI): 51.7%. Top covariate: CNY_USD (8.8%). Diagonal dominance reflects BinaryAttentionBias -- a learned bias that distinguishes same-variable vs cross-variable tokens. Colorscale capped at 60%.

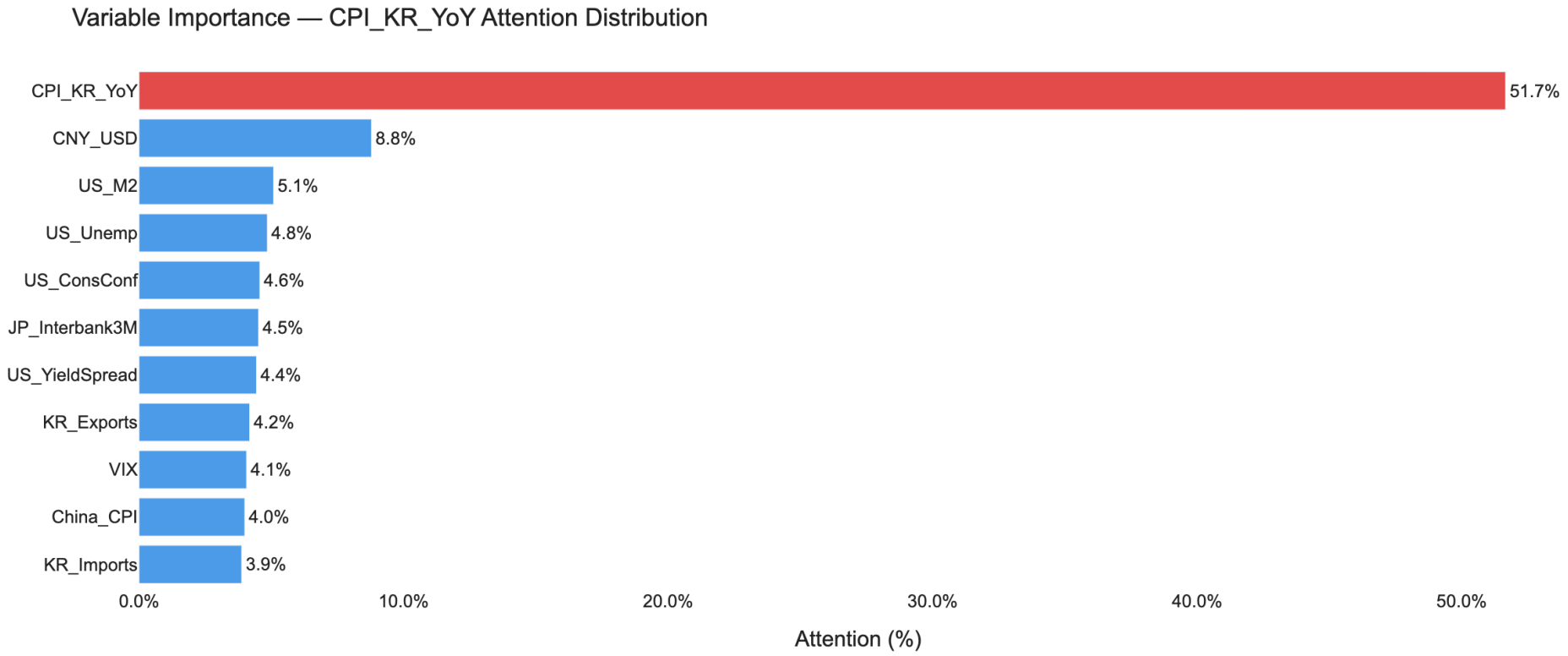
Cross-Variate Attention Matrix (Stage 2, Layer 12)

Query \ Key	CPI_KR_YoY	CNY_USD	US_M2	US_ConsConf	US_Unemp	JP_Interbank3M	US_YieldSpread	KR_Exports	VIX	China_CPI	KR_Imports
CPI_KR_YoY	51.7%	8.8%	5.1%	4.6%	4.8%	4.5%	4.4%	4.2%	4.1%	4.0%	3.9%
CNY_USD	8.9%	47.4%	9.8%	4.6%	4.7%	4.5%	4.4%	4.0%	3.9%	4.1%	3.8%
US_M2	3.3%	7.1%	49.3%	11.5%	4.5%	4.4%	4.5%	3.9%	4.0%	3.8%	3.7%
US_ConsConf	3.6%	3.5%	7.9%	53.1%	8.2%	4.3%	4.3%	3.8%	3.8%	3.9%	3.7%
US_Unemp	3.6%	3.7%	4.6%	7.5%	52.3%	8.0%	4.5%	4.0%	4.0%	3.9%	3.8%
JP_Interbank3M	3.8%	3.7%	4.5%	4.7%	7.0%	53.0%	7.5%	4.0%	4.1%	3.8%	3.9%
US_YieldSpread	3.7%	3.9%	4.6%	4.8%	4.2%	6.6%	52.6%	8.0%	4.0%	3.9%	3.9%
KR_Exports	3.4%	3.5%	4.7%	5.0%	4.8%	4.6%	6.5%	52.8%	6.9%	4.1%	3.8%
VIX	3.4%	3.4%	4.8%	4.9%	4.9%	4.5%	4.7%	5.8%	53.5%	6.3%	3.8%
China_CPI	3.5%	3.6%	4.7%	5.2%	4.8%	4.6%	4.8%	4.1%	5.3%	54.5%	5.0%
KR_Imports	3.6%	3.7%	4.7%	4.9%	4.8%	4.5%	4.6%	4.2%	4.1%	4.9%	56.0%

Attention: 60%, 50%, 40%, 30%, 20%, 10%, 0%

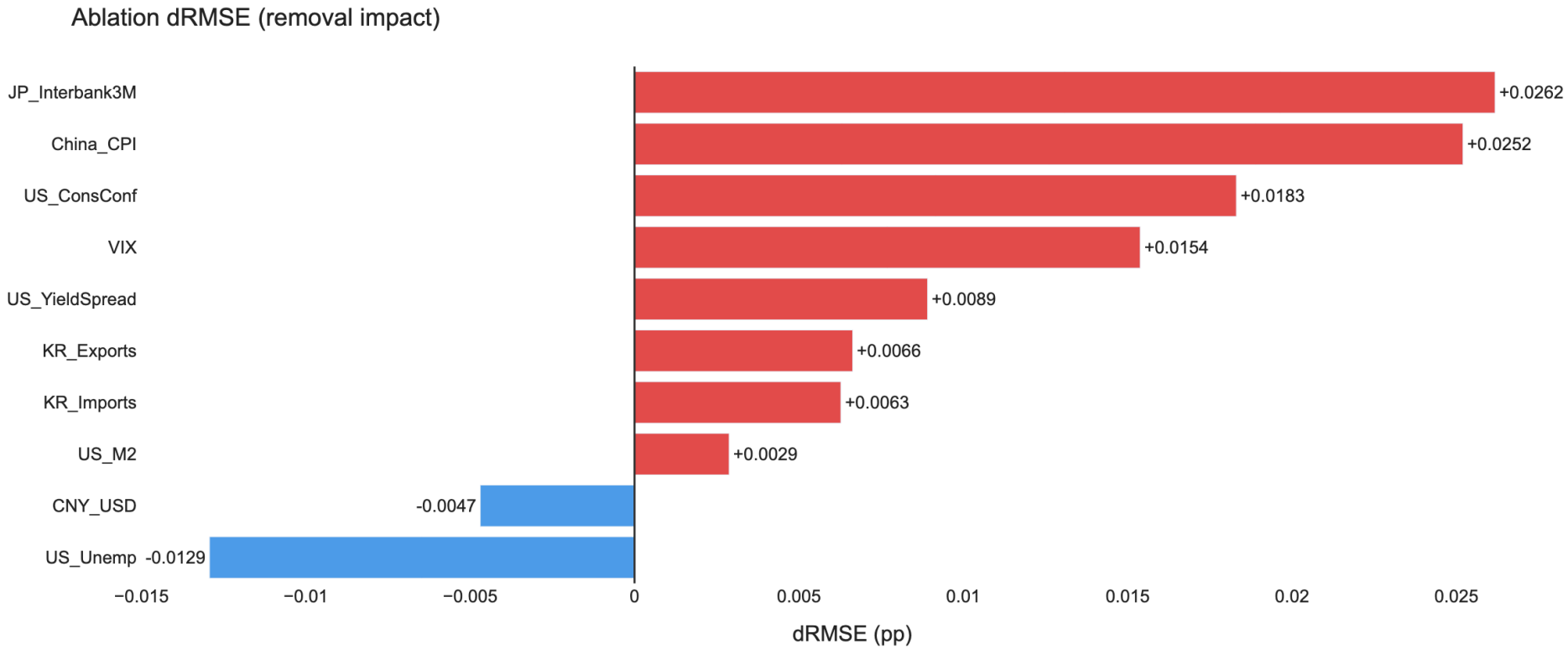
4. Variable Importance

CPI target row from the cross-variate matrix. Self-attention dominates (~52%). Among covariates, attention is distributed relatively evenly (3-5% each), reflecting the model's multivariate information aggregation.



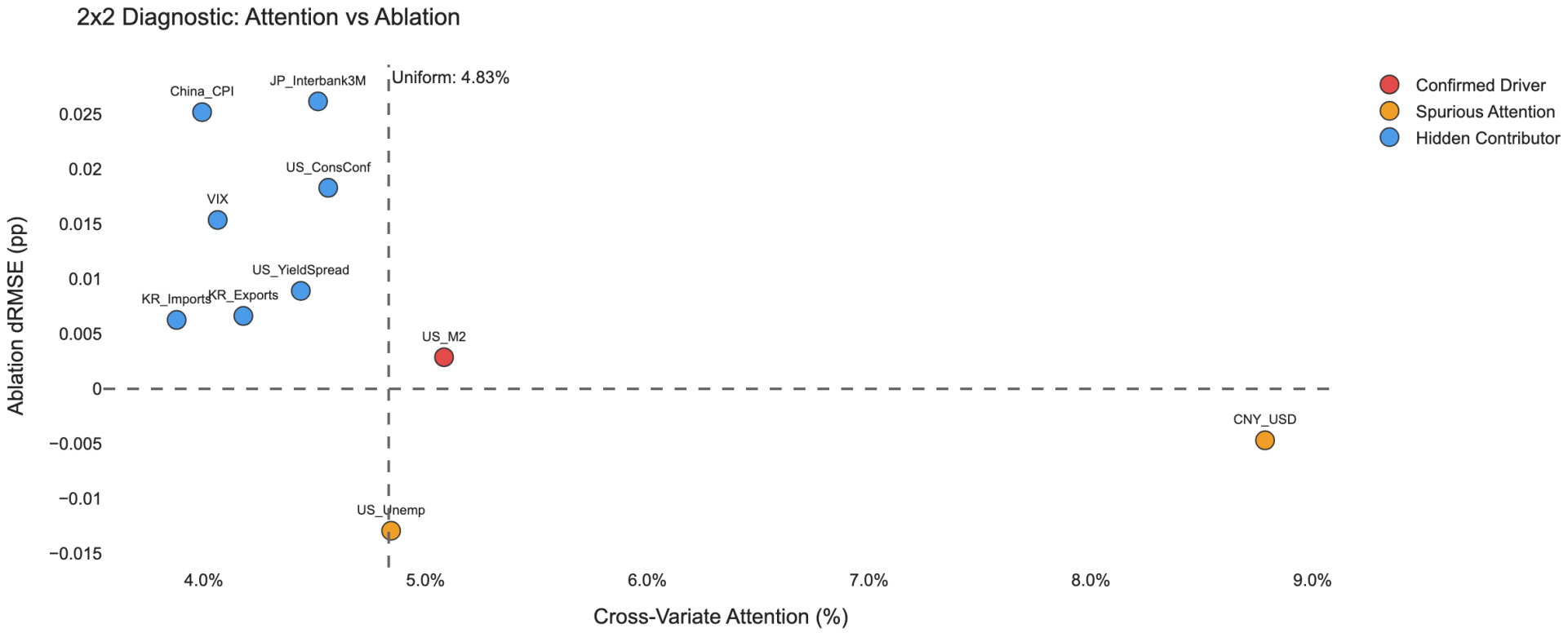
5. Ablation Study

Each variable removed one at a time; dRMSE = RMSE(removed) - RMSE(full). dRMSE > 0: removal hurts forecast (variable contributes). Baseline RMSE: 1.0348pp. Variables with positive dRMSE: 8/10.



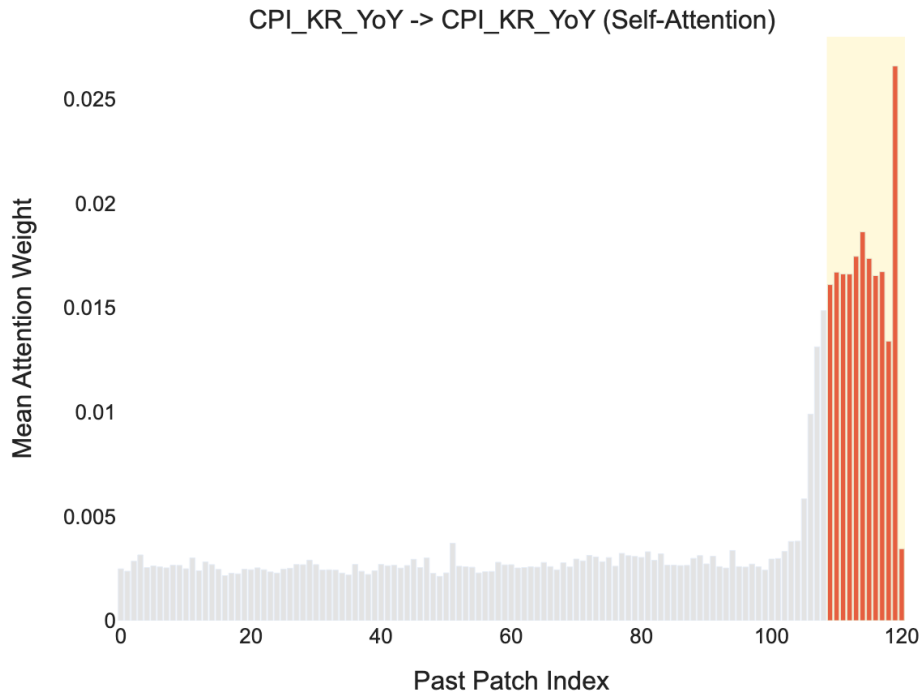
6. 2x2 Diagnostic: Attention vs Ablation

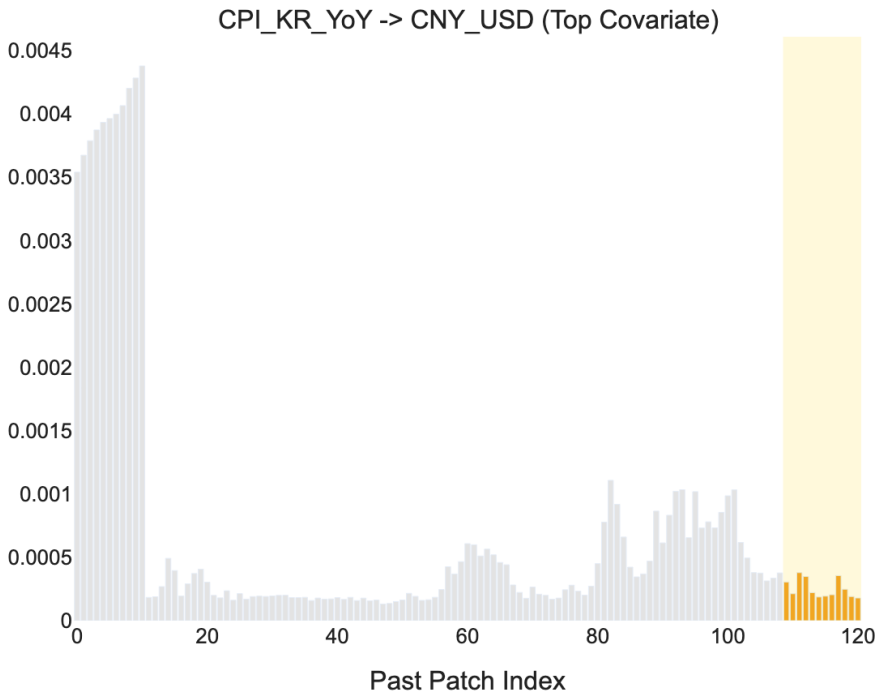
Attention =/= Causal Importance. Vertical line = uniform attention threshold. Confirmed Driver: high attention + positive dRMSE (genuine predictors). Spurious Attention: high attention + low dRMSE (correlated but redundant). Hidden Contributor: low attention + positive dRMSE (unique info not captured by attention).



7a. Temporal Attention: Self vs Top Covariate

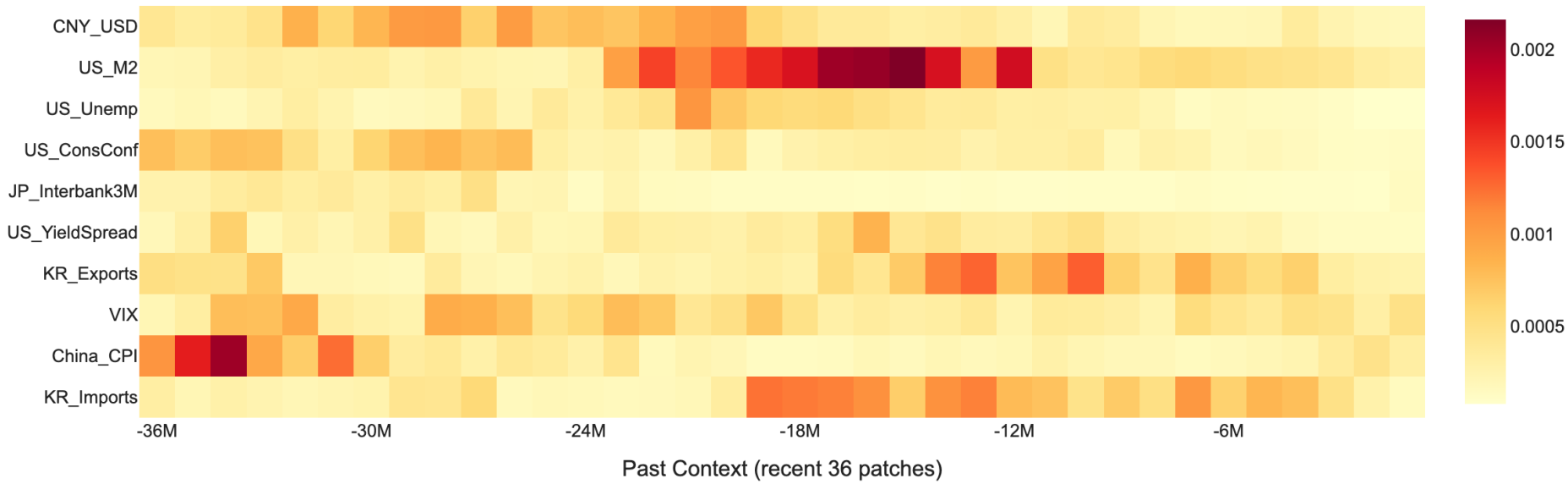
Self-attention (CPI_KR_YoY->CPI_KR_YoY): peak at recent (patch 119) -- natural AR pattern, recent CPI values drive prediction.
Top covariate (CPI_KR_YoY->CNY_USD): peak at distant past (patch 10).
CNY_USD references long-term level (~110 months ago). This explains the Attention-Ablation discrepancy: long-term level reference raises pattern similarity (attention) but may not contribute to forecast accuracy (ablation).





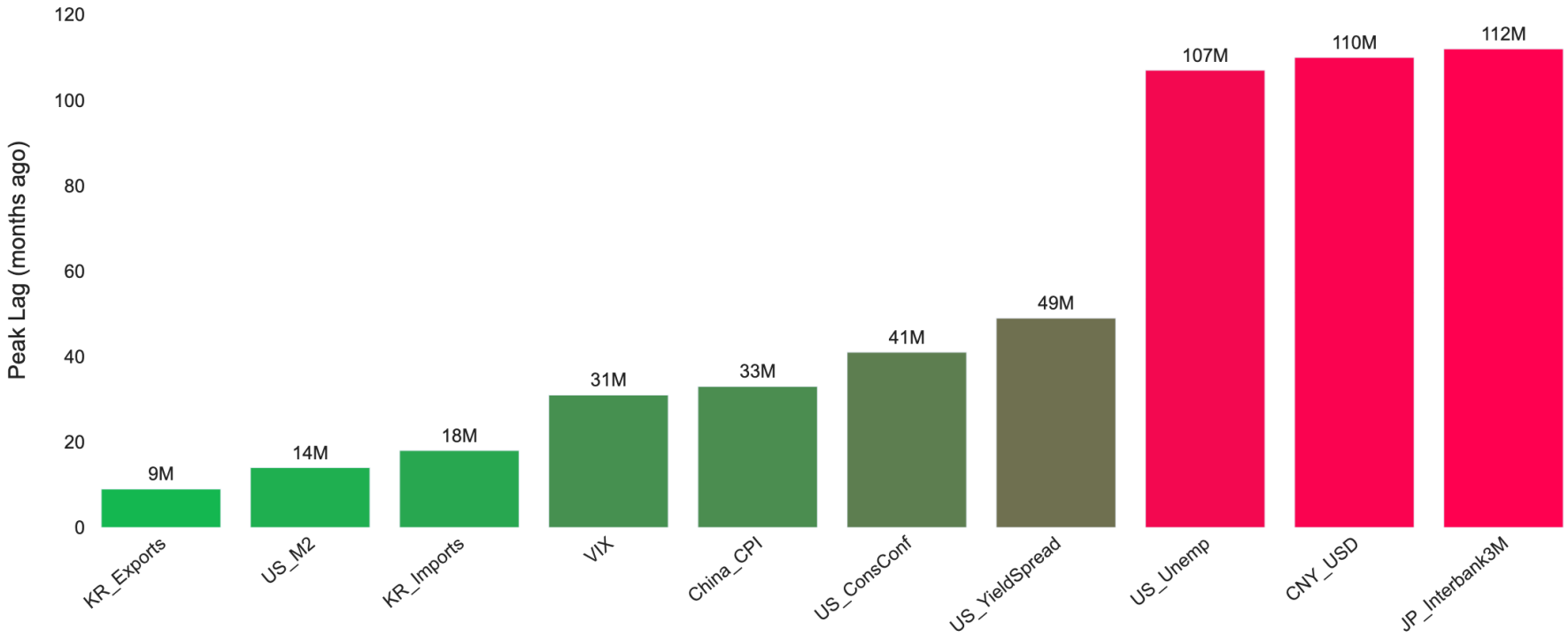
7b. Temporal Attention Heatmap (all covariates)

Each row = one covariate, showing attention weight across the recent 36 patches. Brighter = higher attention. Variables sorted by cross-variate attention rank. Lag structure types: Diffuse: 6, Distant Focus: 3, Mid-range: 1.



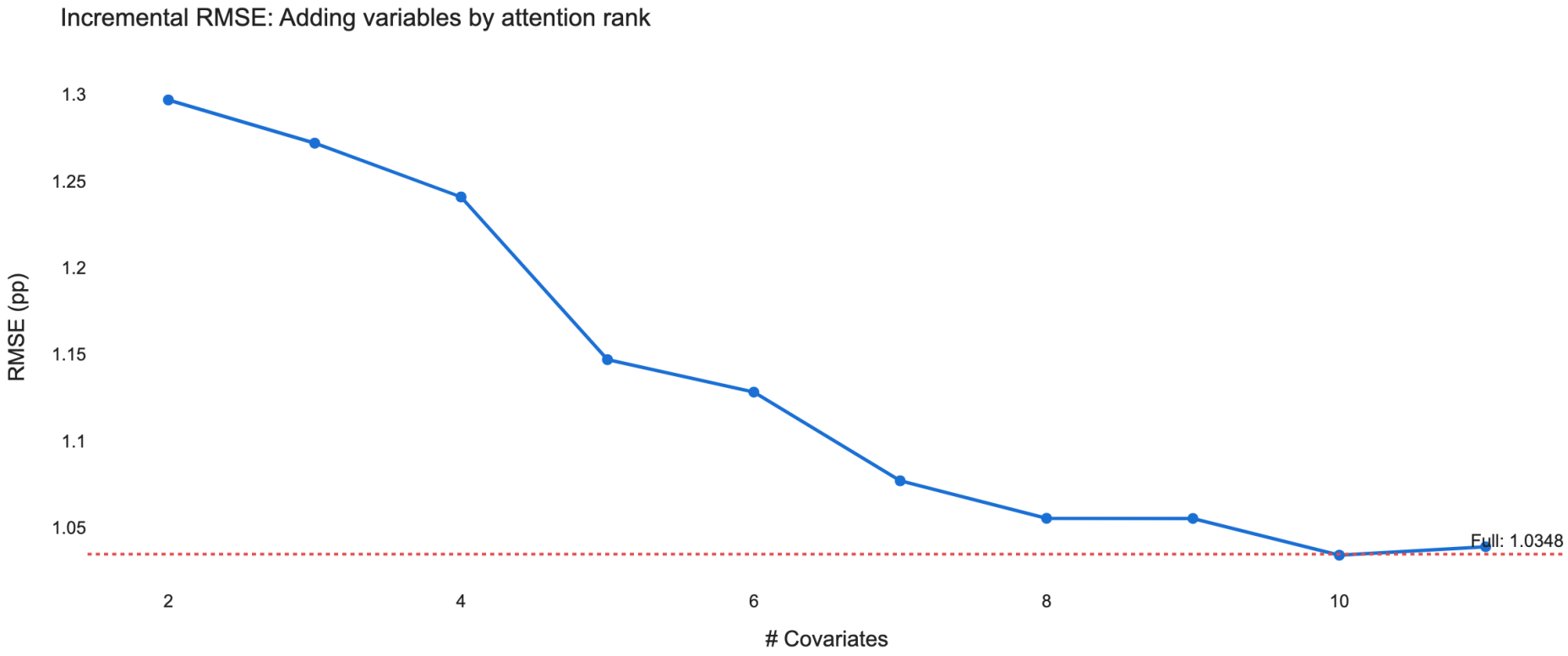
7c. Peak Attention Lag by Variable

Bar height = number of months ago where attention peaks for each variable. Short lag (green) = recent-focused (coincident/short-leading indicators). Long lag (red) = distant-focused (long-term structure reference or artifact).



8. Incremental Analysis

Adding variables one by one in attention-rank order. Best RMSE at 10 covariates (1.0342pp). Full model (11 covariates): 1.0348pp. Diminishing returns after the top variables, suggesting attention ranking effectively identifies the most informative covariates.



9. Summary Table - Attention vs Ablation

#	Variable	Attn Rank	Attn %	dRMSE	Abl Rank	Quadrant
1	JP_Interbank3M	6	4.52%	+0.0262	1	Hidden Contributor
2	China_CPI	8	3.99%	+0.0252	2	Hidden Contributor
3	US_ConsConf	4	4.56%	+0.0183	3	Hidden Contributor
4	VIX	9	4.06%	+0.0154	4	Hidden Contributor
5	US_YieldSpread	5	4.44%	+0.0089	5	Hidden Contributor
6	KR_Exports	7	4.18%	+0.0066	6	Hidden Contributor
7	KR_Imports	10	3.88%	+0.0063	7	Hidden Contributor
8	US_M2	2	5.08%	+0.0029	8	Confirmed Driver
9	CNY_USD	1	8.79%	-0.0047	9	Spurious Attention
10	US_Unemp	3	4.84%	-0.0129	10	Spurious Attention